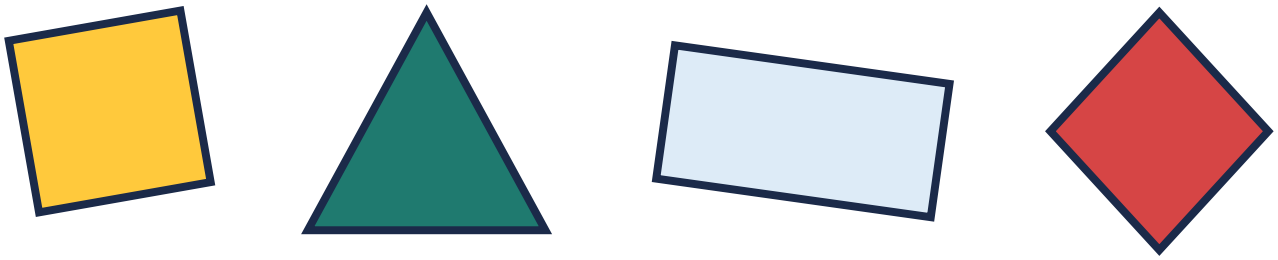
FOR PARENTS • PRIMARY 5 HOLIDAY PROGRAM

# Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



**Notice it. Test it. Explain why.**

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

# 00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

## One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Challenge questions if your child enjoys them.

## The three levels

**Core** is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day.

## What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

# W1 Million Puzzles

LESSON 1 • NUMBERS TO 10 MILLION

## Core

**1** Write 4 030 205 in words.

**2** What is the value of the digit 6 in 3 621 480?

**3** Round 5 482 716 to the nearest thousand.

**4** Work out 2 750 000 ÷ 1000.

## Think Harder

**5** How many 7-digit numbers have digits that add up to 2? List them all.

**6** Put ONE pair of brackets into 2 + 3 × 4 + 5 to make 25. Then put one pair in a different place to make 29.

**7** A whole number rounded to the nearest thousand is 640 000. What is the greatest it could be? The smallest?

### What your child learns

Reading, comparing and rounding numbers up to 10 million, and riddles about large numbers.

### Tip for you

Find a big number in the news, such as a population or a budget. Ask your child to say it in words and round it to the nearest million.

### Ask

"Which digit decides how you round?"

## Challenge

**8** How many 7-digit numbers are there altogether?

**9** Write 1 000 000 as the product of two whole numbers that have no zero digits at all.

# W1 Fraction Feats

LESSON 2 • FRACTIONS: FOUR OPERATIONS

## Core

**1** Write 7 ÷ 9 as a fraction.

**2** Work out 2/3 + 3/4.

**3** Work out 5/6 − 1/4.

**4** Find 3/5 of 15.

## Think Harder

**5** Write 2/3 as the sum of two different unit fractions.

**6** Ben ate 1/3 of a pizza. Ali ate 1/4 of what was left. What fraction of the pizza is left?

**7** 1/2 − 1/3 = 1/6 and 1/3 − 1/4 = 1/12. Use the pattern to work out 1/4 − 1/5 and 1/9 − 1/10.

### What your child learns

Adding, subtracting, multiplying and dividing fractions, using bar models and pictures.

### Tip for you

Share a block of chocolate or a cake. Ask what is left after you eat 1/3 and then half of what remains.

### Ask

"Can you show it with a bar?"

## Challenge

**8** Write 1 as the sum of three different unit fractions. Find all the ways.

**9** Work out $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \ldots \times \frac{99}{100}$.

# W2 Before and After

LESSON 3 • FRACTION WORD PROBLEMS

## Core

**1** A baker made some buns. She sold 2/5 of them in the morning and 1/3 of the remaining buns in the afternoon. She had 120 buns left. How many buns did she make?

**2** Ben is 10 years old and his father is 38. In how many years will Ben's age be 1/2 of his father's age?

**3** Siti had $60. She spent 1/4 of it on a book and 1/3 of the remainder on a T-shirt. How much money did she have left?

**4** Ali had 26 stamps and Ravi had 34. How many stamps must Ravi give Ali so that they have the same number?

## Think Harder

**5** Ravi spent 1/4 of his money on Monday and 2/5 of the remainder on Tuesday. He spent $66 altogether. How much money did he have at first?

**6** A box had red pens and blue pens. 2/3 of the pens were blue. After 12 blue pens were taken out, 1/2 of the pens were blue. How many pens were in the box at first?

### What your child learns

Fraction word problems: before-and-after situations and 'fraction of the remainder', solved with bar models.

### Tip for you

Make up a story: "I spent 1/4 of my money, then 1/2 of the rest, and had $12 left." Let your child draw it and work backwards.

### Ask

"What is the whole here, and what is it now?"

**7** Mrs Tan is 41 years old and her son is 9. How many years ago was her son's age 1/5 of her age?

## Challenge

**8** Ali, Ben and Mei shared $180. Ali took 1/3 of the money, Ben took 1/4 of what was left and Mei took the rest. How much more did Mei get than Ali?

**9** Mei had some money. She spent $12, then spent 1/2 of what was left, then spent $8. She had $10 left. How much did she have at first?

# W2 Triangle Areas

LESSON 4 • AREA OF TRIANGLE

## Core

**1** A triangle has a base of 12 cm and a height of 5 cm. What is its area?

**2** A right-angled triangle has sides of 8 cm, 15 cm and 17 cm. What is its area?

**3** A triangle has an area of 36 cm² and a height of 8 cm. How long is its base?

**4** A rectangle 9 cm by 6 cm is cut along a diagonal into two triangles. What is the area of each triangle?

## Think Harder

**5** A triangle has an area of 45 cm². Its base is 10 cm. What is its height?

**6** A right-angled triangle has sides of 12 cm, 16 cm and 20 cm. What is its area?

### What your child learns

Area of a triangle: base and height, why the area is half of a rectangle, and triangles in composite figures.

### Tip for you

Cut a rectangle of paper in half along the diagonal. Ask why each triangle is half of the rectangle.

### Ask

"Which side is the base, and where is the height?"

**7** A square has sides of 10 cm. A triangle is drawn with one side of the square as its base and its top corner on the opposite side. What is the area of the triangle?

## Challenge

**8** Triangle ABC has an area of 60 $cm^2$. D is the middle point of BC. What is the area of triangle ABD?

**9** A rectangle is 12 cm by 8 cm. P is the middle of the top side and Q is the middle of the right side. What is the area of the triangle joining P, Q and the bottom-left corner?

# W3 LESSON 5 • VOLUME Volume Vault

## Core

**1** A cuboid is 9 cm long, 4 cm wide and 5 cm high. What is its volume?

**2** A cube has edges of 6 cm. How many 1-cm cubes are needed to build it?

**3** A tank 40 cm long, 20 cm wide and 30 cm high is half full of water. How many litres of water are in it?

**4** A cuboid has a volume of 240 $cm^3$. Its base is 8 cm by 6 cm. How high is it?

## Think Harder

**5** How many cubes with 2-cm edges can be packed in neat layers into a box 9 cm long, 6 cm wide and 5 cm high?

**6** A tank is 1/3 full of water. After 12 ℓ more is poured in, it is 3/5 full. How many litres does the tank hold when full?

### What your child learns

Volume of cuboids, water in tanks, stones that sink, and painted-cube puzzles.

### Tip for you

Fill a box with sugar cubes or blocks. Ask for the number of cubes before counting, then check.

### Ask

"How many cubes in one layer, and how many layers?"

**7** A cuboid has a square base and a height of 5 cm. Its volume is 245 $cm^3$. How long is each side of the base?

## Challenge

**8** A big cube is painted on the outside and cut into small cubes of the same size. 96 small cubes have exactly one painted face. How many small cubes are there altogether?

**9** How many different cuboids can be built with exactly 64 one-centimetre cubes, using all of them each time?

# W3 LESSON 6 • RATE Rate Racers

## Core

**1** 5 kg of flour cost $8.50. How much do 3 kg cost?

**2** A printer prints 24 pages a minute. How long does it take to print 360 pages?

**3** Which is the better buy: 4 rolls for $3.40 or 6 rolls for $4.98?

**4** A tap fills a 120 ℓ tank at 8 ℓ a minute. How long does it take?

## Think Harder

**5** A tap fills a tank in 12 minutes at 15 ℓ a minute. When a second tap is also turned on, the tank fills in 9 minutes. How many litres a minute does the second tap fill?

**6** Eggs are sold in boxes of 10 for $3.30 and boxes of 6 for $2.10. Siti needs at least 28 eggs. What is the least she can pay?

### What your child learns

Rate: the amount for one, better buys, taps filling a tank, and working out a charge 'or part of' an hour.

### Tip for you

At the supermarket, compare two pack sizes of the same item. Which costs less for one?

### Ask

"What does it cost for just one?"

**7** A car park charges $1.50 for the first hour and $1 for every half hour or part of a half hour after that. How much does it cost to park from 10.15 a.m. to 2.50 p.m.?

## Challenge

**8** Printer P prints a job in 6 minutes. Printer Q prints the same job in 3 minutes. How long do they take working together?

**9** Siti buys pens at 3 for $2 and sells them at 2 for $2. How many pens must she buy and sell to make a profit of $10?

# W4 Percentage Puzzles

LESSON 7 • PERCENTAGE

## Core

**1** Find 10% of $360 and 15% of $360.

**2** A book costs $45. It is sold at a 20% discount. What is the sale price?

**3** A meal costs $30 before GST. GST is 9%. How much is the meal with GST?

**4** 24 of the 40 pupils in a class walk to school. What percentage walk to school?

## Think Harder

**5** At a hawker centre, a plate of chicken rice went up from $4 to $5. A month later, it went back down from $5 to $4. (a) Find the percentage increase from $4 to $5. (b) Find the percentage decrease from $5 to $4. (c) Both changes are $1. Why are the two percentages different? Hint: percentage change = the change out of the amount you START with.

### What your child learns

Percentage as a fraction of 100, finding a part, discount, GST and percentage change.

### Tip for you

Look at a sale tag. Ask for the price after 20% off, then check with a calculator.

### Ask

"Percentage of what? What is the 100%?"

**6** A box of cereal costs $15 in two shops. Shop X: "Buy 3, get 1 free!" Shop Y: "30% off every box!" Mei buys 4 boxes. (a) How much does she pay at each shop? Which is the better buy, and by how much? (b) Shop X's deal is really how many percent off?

**7** Siti says, "15% of $80 is $8 + $4 = $12." Explain her method. Use it to find 15% of $46.

## Challenge

**8** A $100 jacket goes up by 20%, then the new price goes down by 20%. What is the final price? Is it a percentage increase or decrease from $100, and how much?

**9** After a 20% discount, a shirt costs $40. Ben says the price before the discount was $48. Is he right? Find the price before the discount.

# W4 Angle Chasers

LESSON 8 • ANGLES, TRIANGLES AND QUADRILATERALS

## Core

**1** Two angles on a straight line. One is 127°. What is the other?

**2** Four angles meet at a point. Three are 150°, 90° and 80°. What is the fourth?

**3** Two angles of a triangle are 48° and 67°. What is the third angle?

**4** An isosceles triangle has a top angle of 30°. What is each base angle?

## Think Harder

**5** Two straight lines cross. One of the four angles is 38°. What are the other three? How many different sizes are there?

**6** An isosceles triangle has an angle of 110°. Find the other two angles. Why is there only one answer?

### What your child learns

Angle rules for triangles and quadrilaterals, finding unknown angles and giving a reason for each step.

### Tip for you

Tear off the three corners of a paper triangle and put them side by side. What do they make?

### Ask

"Which angle rule did you use, and why?"

**7** In trapezium ABCD, AB is parallel to DC. Angle DAB is 84° and angle ABC is 97°. Find angle ADC and angle BCD.

## Challenge

**8** Two identical right-angled triangles with angles 30°, 60° and 90° are joined with their shorter sides (the sides opposite the 30° angles) touching, with the right angles together, to make one big triangle. What are the angles of the big triangle?

**9** Three angles of a four-sided shape are 95°, 110° and 85°. What is the fourth angle? Could the shape be a parallelogram?

# W5 Pattern Generalisers

LESSON 9 • PATTERNS AND NUMBER SENSE

## Core

**1** 5, 9, 13, 17, … What is the 10th number in this pattern?

**2** Matchstick squares in a row use 4, 7, 10, … sticks. How many sticks are needed for 8 squares?

**3** 2, 6, 10, 14, … Is 50 in this pattern? Explain.

**4** There are 15 bicycles and tricycles in a shed. They have 37 wheels altogether. How many tricycles are there?

## Think Harder

**5** Beads repeat in the order red, red, blue, yellow, yellow, yellow. What colour is the 50th bead? How many yellow beads are in the first 100?

**6** 40 square tables in one long row seat $2 \times 40 + 2 = 82$ people. How many can sit if the tables are put in 8 rows of 5 tables instead?

### What your child learns

Patterns and number sense: finding the rule, predicting far-away terms, and checking with a table.

### Tip for you

Make a matchstick pattern. Ask for the number of sticks in the 20th figure before you build it.

### Ask

"Can you test your rule on the next figure?"

**7** A farm has chickens and goats. There are 35 heads and 94 legs. How many goats are there?

## Challenge

**8** A staircase is built from squares: step 1 is 1 square, step 2 adds a column of 2 (3 squares altogether), step 3 adds a column of 3 (6 squares). How many squares are there in the staircase with 10 steps?

**9** Matchstick triangles in a row use 3, 5, 7, … sticks. Which figure uses exactly 99 sticks?

# W5 LESSON 10 • MIXED HEURISTICS Challenge Day

## Core

**1** A number is multiplied by 4, then 6 is taken away, and the answer is 50. What was the number?

**2** Ali has 1/2 of his money left after buying a book for $9. How much did he have at first?

**3** Two numbers add up to 40 and the bigger is 8 more than the smaller. What are the two numbers?

**4** Ravi, Siti and Ben each have a different pet: a rabbit, a hamster and a bird. Ravi has no bird and no hamster. Siti has no hamster. Who has the bird?

## Think Harder

**5** Siti spent 2/5 of her money, then 1/3 of what was left. She had $24 left at the end. How much did she start with?

**6** How many 3-digit numbers have digits that add up to 5? List them all in order.

### What your child learns

Choosing a strategy: guess and check, working backwards, making a table and drawing a model.

### Tip for you

When your child is stuck, ask "Can you draw it?" or "Can you try a number?" before giving any other hint.

### Ask

"How did you start? What would you try next time?"

**7** The sum of two consecutive numbers (numbers that are next to each other, like 40 and 41) is 145. What are the two numbers?

## Challenge

**8** At the first bus stop, 1/3 of the passengers got off. At the second stop, half of those left got off and 3 got on. There are now 12 passengers. How many were on the bus at first?

**9** How many rectangles of all sizes are there in a 1 by 5 strip of 5 small squares?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

# Answers

## Week 1 • Lesson 1: Million Puzzles

1. Four million, thirty thousand, two hundred and five.
2. 600 000
3. 5 483 000
4. 2750
5. 7 numbers: 1 000 001, 1 000 010, 1 000 100, 1 001 000, 1 010 000, 1 100 000 and 2 000 000.
6. (2 + 3) × 4 + 5 = 25 and 2 + 3 × (4 + 5) = 29. (29 is the biggest you can make with one pair.)
7. Greatest 640 499, smallest 639 500.
8. 9 000 000. They go from 1 000 000 to 9 999 999, and 9 999 999 − 1 000 000 + 1 = 9 000 000.
9. 64 × 15 625. (1 000 000 = 10 × 10 × 10 × 10 × 10 × 10; put all the 2s together to make 64 and all the 5s together to make 15 625.)

## Week 1 • Lesson 2: Fraction Feats

1. 7/9
2. 17/12 = 1 5/12
3. 7/12
4. 9
5. 1/2 + 1/6 (the only way).
6. 1/2. 2/3 was left after Ben; Ali ate 1/4 of it, so 3/4 of 2/3 = 1/2 is left.
7. 1/20 and 1/90 (the bottom numbers multiply: 4 × 5 and 9 × 10).
8. Only one way: 1/2 + 1/3 + 1/6. (The biggest part must be 1/2, and then 1/2 = 1/3 + 1/6 is the only split into two different unit fractions.)
9. 1/100. Each top number cancels the bottom number before it, leaving 1 on top and 100 at the bottom.

## Week 2 • Lesson 3: Before and After

1. 300 (left = 2/3 of 3/5 = 2/5 of all the buns; 2/5 = 120, so 1/5 = 60). She sold 60 in the afternoon.
2. 18 years (the difference, 28, is 1 unit; Ben will be 28 and his father 56)
3. $30 (book $15, remainder $45, T-shirt $15)
4. 4 (the total is 60, so 30 each)
5. $120 (20 units: Monday 5 units, Tuesday 6 units; 11 units = $66, so 1 unit = $6)
6. 36 (the red pens did not change: red 1 unit, blue 2 units at first, 1 unit after; 1 unit = 12)
7. 1 year ago (the difference, 32, is 4 units; 1 unit = 8: son 8, Mrs Tan 40)
8. $30 (Ali $60, Ben $30, Mei $90)
9. $48 (work backwards: 10 + 8 = 18; 18 × 2 = 36; 36 + 12 = 48)

## Week 2 • Lesson 4: Triangle Areas

1. 30 cm²
2. 60 cm² (1/2 × 8 × 15; 17 cm is the slanting side)
3. 9 cm (1/2 × 9 × 8 = 36)
4. 27 cm²
5. 9 cm
6. 96 cm² (1/2 × 12 × 16)
7. 50 cm²
8. 30 cm² (half the base, same height)
9. 36 cm² (96 − 24 − 12 − 24)

## Week 3 • Lesson 5: Volume Vault

1. 180 cm³
2. 216 (6 × 6 × 6)
3. 12 ℓ (40 × 20 × 15 = 12 000 cm³)
4. 5 cm (240 ÷ 48)
5. 24 (4 along, 3 across, 2 layers; not 270 ÷ 8, because the cubes cannot be cut)
6. 45 ℓ (3/5 – 1/3 = 4/15 of the tank is 12 ℓ, so 1/15 is 3 ℓ)
7. 7 cm (base area 245 ÷ 5 = 49 cm², and 7 × 7 = 49)
8. 216 (each face has 96 ÷ 6 = 16 = 4 × 4 one-face cubes, so the cube is 6 cubes along each edge: 6 × 6 × 6)
9. 7 (1 × 1 × 64, 1 × 2 × 32, 1 × 4 × 16, 1 × 8 × 8, 2 × 2 × 16, 2 × 4 × 8, 4 × 4 × 4)

## Week 3 • Lesson 6: Rate Racers

1. $5.10 ($1.70 a kg)
2. 15 minutes
3. 6 rolls for $4.98 (83¢ a roll, against 85¢)
4. 15 minutes
5. 5 ℓ a minute (the tank holds 180 ℓ; 180 ÷ 9 = 20 ℓ a minute together; 20 – 15 = 5)
6. $9.60 (one box of 10 and three boxes of 6 make 28 eggs; three boxes of 10 cost $9.90)
7. $9.50 (4 h 35 min: first hour, then 3 h 35 min is 8 half hours or parts)
8. 2 minutes (pretend the job is 6 pages: P prints 1 page a minute, Q 2 pages, together 3 a minute)
9. 30 pens (6 pens cost $4 and sell for $6: $2 profit for every 6 pens)

## Week 4 • Lesson 7: Percentage Puzzles

1. $36 and $54 ($36 + $18)
2. $36 (20% of $45 = $9)
3. $32.70 (9% of $30 = $2.70)
4. 60% (24 out of 40 = 60 out of 100)
5. (a) 25% ($1 out of $4) (b) 20% ($1 out of $5) (c) Each change is compared with its own starting amount: $1 out of $4 is a bigger part than $1 out of $5. (Percentage change is a P6 skill: a preview.)
6. (a) Shop X $45 (she pays for 3 boxes); Shop Y $42 (30% of $60 = $18 off). Shop Y is the better buy, by $3. (b) 25% off (1 box free out of 4, $15 off $60)
7. She found 10% ($8) and 5% (half of 10%, $4) and added them. 15% of $46 = $4.60 + $2.30 = $6.90.
8. $96 ($120, then 20% of $120 = $24 off). A 4% decrease.
9. No. $40 is 80% of the old price, so 10% is $5 and 100% is $50. (Ben added 20% of $40, but the 20% was of the old price.)

## Week 4 • Lesson 8: Angle Chasers

1. 53° (180° – 127°)
2. 40° (360° – 320°)
3. 65° (180° – 115°)
4. 75° ((180° – 30°) ÷ 2)
5. 142°, 38°, 142°. There are two different sizes: 38° and 142°.
6. 35° and 35°. 110° cannot be a base angle (two of them would be 220°), so it is the top angle and 180° – 110° = 70° is shared by two equal base angles.
7. ADC = 96° and BCD = 83° (angles between the parallel sides add up to 180°). Check: 84° + 97° + 96° + 83° = 360°.
8. 30°, 30°, 120°. The two right angles make a straight line, and the two 60° angles together make 120°.
9. 70° (360° – 290°). It cannot be a parallelogram: the angles 95°, 110°, 85° and 70° have no equal opposite pairs.

## Week 5 • Lesson 9: Pattern Generalisers

1. 41 (5 + 9 × 4: there are 9 jumps of 4 from the 1st number to the 10th).
2. 25 (3 × 8 + 1).
3. Yes. 50 − 2 = 48 and 48 ÷ 4 = 12 jumps, so 50 is the 13th number.
4. 7 tricycles (suppose all 15 are bicycles: 30 wheels; 7 wheels short; each tricycle adds 1 more wheel). 8 bicycles. Check: 16 + 21 = 37.
5. Red (50 = 8 × 6 + 2, so it is the 2nd bead of a group). 49 yellow beads: 16 full groups of 6 give 96 beads and 48 yellow, and beads 97 to 100 are red, red, blue, yellow.
6. 96 (each row of 5 seats 12, and 8 × 12 = 96). One long row seats only 82. Splitting the tables gives more end seats.
7. 12 goats and 23 chickens (all chickens would have 70 legs; 24 legs short; each goat adds 2 legs). Check: 48 + 46 = 94.
8. 55 (1 + 2 + 3 + … + 10). Two copies of the staircase make a 10 by 11 rectangle: 110 ÷ 2 = 55.
9. Figure 49 (sticks = 2 × figure number + 1; 99 − 1 = 98, 98 ÷ 2 = 49).

## Week 5 • Lesson 10: Challenge Day

1. 14 (work backwards: 50 + 6 = 56, 56 ÷ 4 = 14).
2. $18 ($9 is the other half).
3. 24 and 16 (take away the 8: 32, share: 16 and 16, then add 8 back to one: 24 and 16).
4. Siti. Ravi has the rabbit (no bird, no hamster). Siti is not the hamster, so she has the bird, and Ben has the hamster.
5. $60. Work backwards: $24 is 2/3 of what was left, so that was $36; $36 is 3/5 of her money, so 1/5 is $12 and she started with $60. Check: 60 − 24 = 36, 36 − 12 = 24.
6. 15 numbers: 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500.
7. 72 and 73 (145 is 1 more than twice 72).
8. 27. Work backwards: 12 − 3 = 9; 9 is half of what was left, so 18 were left after stop 1; 18 is 2/3 of the passengers, so 1/3 is 9 and there were 27 at first.
9. 15 (5 of length 1, 4 of length 2, 3 of length 3, 2 of length 4 and 1 of length 5: 5 + 4 + 3 + 2 + 1).